

Optoelectronics: An Optimistic Choice?

The market for optical sector is huge in India but there are only a few companies that generate employment opportunities. Let's find out the scope for a career in this domain, the kind of opportunities available in India and the companies that one can look out for



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Optoelectronics is the science that deals with designing devices that can detect or emit light in any part of the spectrum. It is used in a variety of different fields like communication, warfare and consumer applications. Though the market for optical sector is huge in India, there are not enough firms to cater to the whole of it.

Bijoy Krishna Das, associate professor, Department of Electrical Engineering, Indian Institute of Technology (IIT) Madras, informs, "In developed countries, optical technologies are widely used for system development in the areas of communications, distributed sensing, biomedical applications, industrial manufacturing, display and lighting, defence environments, etc. Because of its huge population and recent economic growth, India

has also a gigantic consumer market for those optical components and systems."

"Slowly but steadily, India has started implementing the ambitious fibre-to-the-home (FTTH) project and RF photonics in defence applications," Das adds.

In FTTH, the broadband network architecture utilises optical fibre by replacing all the metallic cables up to the outside wall of a home.

Of late, the government of India has

recognised the potential of optical communication in general and its suitability for broadband Internet services in particular.

"The government is also encouraging photovoltaic as an alternative for generating energy. Biomedical optics including microscopy, endoscopy and skin treatment is another major application area of optoelectronics. Optical computing and new material for sensors and devices are the other major areas where optoelectronics will find extensive usage," says Prof. Naresh Grover, director, Faculty of Engineering, Manav Rachna International University.

Therefore India, with a large population and vast geographical area, has a massive potential for growth in this sector.

According to Pankaj Mittal, director, ARCHELONS, the optical industry in India is currently enjoying a

healthy boom. He reasons, "Increasing westernisation in the last decade or so, coupled with the availability of products matching international standards, and spurred by cable television and social media, has made the optical sector thrive in all its glory. Owing to the growing consumer awareness, and the proportional growing income levels, the Indian consumers' demand for high-quality premium products has seen a dramatic rise like never before."

"Clever marketing strategies assure that the products are now increasingly noticeable. Huge demands in the communication and display sectors have spearheaded the growth in the optical industry," Mittal adds.

Scope

"There are a few companies that exist in India but they are mostly engaged in assembling the optical systems and electronics. They hardly generate any employment opportunities," explains Das.

However, he believes, "Semiconductor giants like Intel and IBM have design houses that offer reasonable job opportunities for optics and optoelectronics engineers. KLA Tencor also offers jobs for optical engineers."

Apart from these, there are many small-scale companies that absorb a major chunk of optics experts across the country.

Prof. Grover says, "At present, the opportunities are limited but in the coming years, there will be an ample scope for jobs in this sector due to growth in the communication sector, healthcare industry, and new materials for sensors and devices needed in the defence sector."

Mittal believes that India has good